Monash University: Assessment Cover Sheet

Student name	Mostafa	Annur	
School/Campus		**Student's I.D. number**	31417094
Unit name	FIT3179 Data visualisation - S2 2023 MUM		
Lecturer's name		**Tutor's name**	
Assignment name	Data Visualisation II Report	**Group Assignment: No** **Note, each student must attach a coversheet**	

Lab/Tute Class:	**Lab/Tute Time:**	**Word Count:**
Due date: 15-10-2023	**Submit Date:**	**Extension granted** ◻

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github: https://github.com/AnnurMostafa01/FIT3179-A2.git

Data Visualisation II

Volcanic Eruptions in the Holocene Period

By: Annur Mostafa
Date: 09/06/2023

number of words: 588

URL of visualisation:https://annurmostafa01.github.io/FIT3179-A2/

Github: https://github.com/AnnurMostafa01/FIT3179-A2.git

Domain?

The topic of interest here is the volcanic eruptions during the Holocene period.The dataset includes the volcano eruptions that took place on Earth over the past 10,000 years. The volcanoes are said to be still active.

Why?

The target audience is of a broad scope as the topic can be merrily treated as a learning subject matter for many or for geologists who have a knack for gaining knowledge about volcanic eruptions.

Who?

Visualisation for the average Australian or Malaysian.

What?

The dataset includes the names, locations, types and features of above 1,500 volcanoes that had erupted in the last 10,000 years or are active till now. The dataset has been retrieved by the Smithsonian Institution's Global Volcanism Program (GVP).

The data set has been extracted from the site kaggle.com.

Url: https://www.kaggle.com/datasets/smithsonian/volcanic-eruptions/data

Dot map

Why?

A dot map is used here where the points depict the volcano eruptions with hue used to represent the elevation in each geographic point. Alternatively, proportional symbol map is not used because size used as a channel can be disastrous as the scalability of the elevation can become difficult in most areas. A choropleth map is also not a wise choice as volcanoes do not appear everywhere which is the same for bin maps.

How?

-Marks: points

Bar Chart

Why?

From the idiom chosen the readers will be able to get insight regarding the volcanic eruptions that took place in different regions of the Earth and their highest and lowest elevations in respective regions. With South America with the highest elevation, and Hawaii and the Pacific Ocean with the lowest.

How?

-Marks: Lines(Bars)

-Channels:

- Length to visualise the elevation
- Ordered by their value of elevation
- hue

Scatter Plot

Why?

The idiom of choice here lets the readers find the trends between the quantitative values, elevation in metres and type of volcanic eruptions. For each type of volcanic eruptions, readers can gain insight into the highest and lowest elevation yet recorded in the Holocene period.

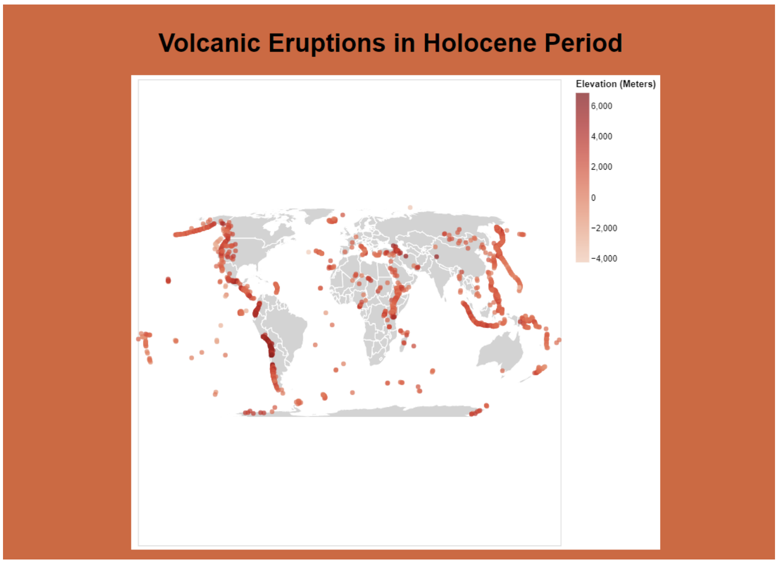
How?

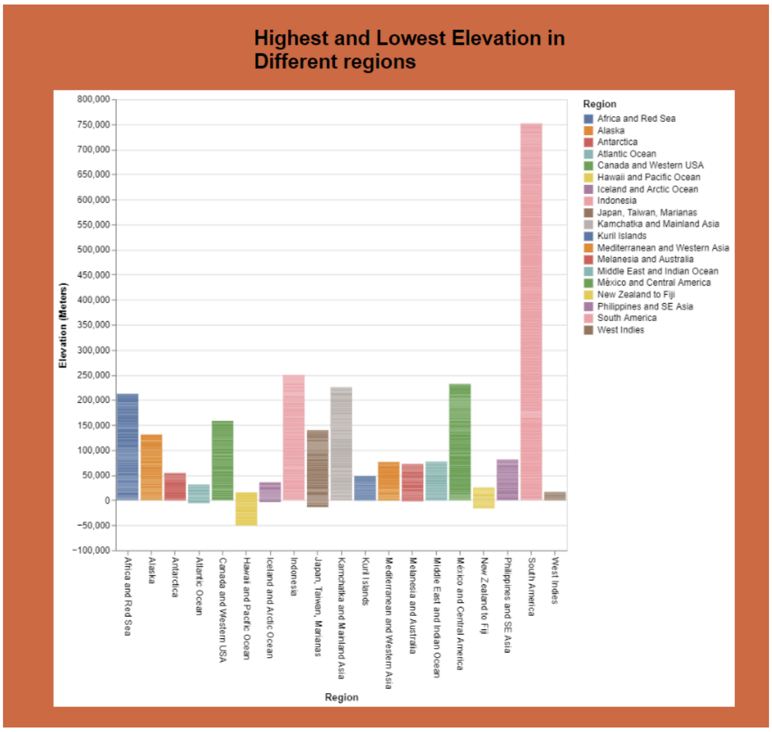
-Marks: points

-Channels:

- vertical positions
- hue

Visualisation





Dot map: The dot map shows the volcanic eruptions that took place around the world during the Holocene period using the coordinates.

Bar chart: Each bar of the bar chart shows the elevation in each region.

Scatterplot: Each point in the scatter plot shows the elevation of each type of volcanic eruptions.

Design

i. Layout:

The visualisation has been properly aligned with an invisible frame around it. The idioms touch the invisible boundary. Few sight lines are used for an easy read. White spaces are also used between each idiom and paragraphs.

ii. Colour:

Two colour palettes are used. One to depict the volcanoes to their elevation, darker the hue higher the elevation vice versa. As for the idioms, different hues are used to represent different nominal data which are, regions to elevations and eruption type to elevations. The readers can thus differentiate between each of them.

iii. Figure-ground:

The marks mostly chosen for each idiom are rather bold and non overlapping so the hierarchy is seen clearly within the idioms even though they are not coinciding with each other.

iv. Typography:

The headings for the idioms are in bold. Whilst other texts are left in light. This will help the readers pay extra attention to the more important parts of the visualisation.

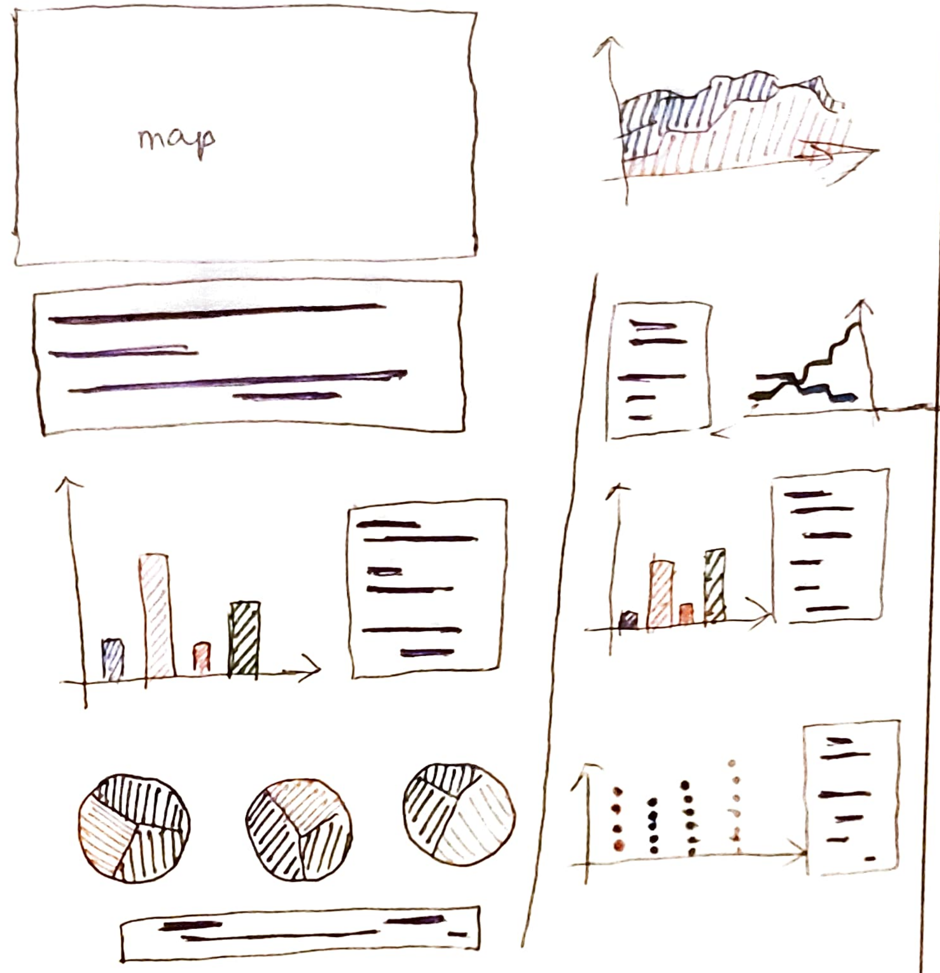
v. Storytelling:

The readers are consciously guided to read from top to bottom with each paragraph attached to their respective idiom.

Appendix

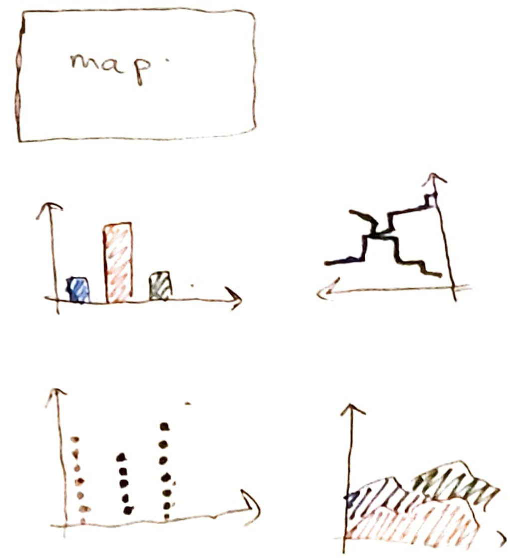
Five Design Sheet methodology

I have gone with sheet 4 as my primary design layout

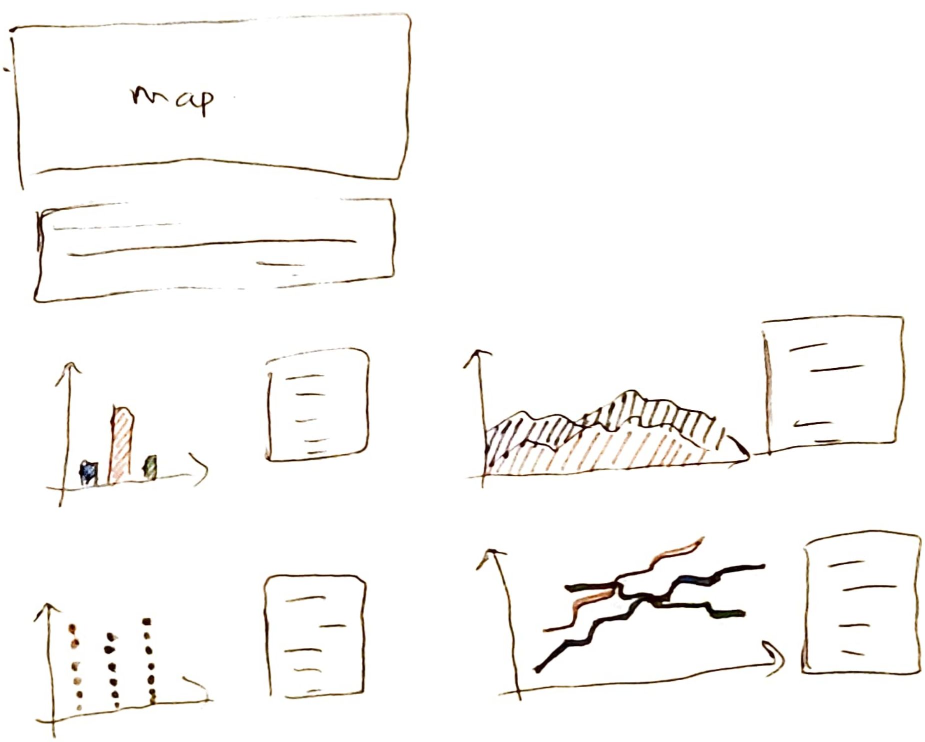


IDEAS

1. Area chart
2. line chart
3. bar chart (interactive)
4. pie chart
5. Scatterplot
6. map (interactive)



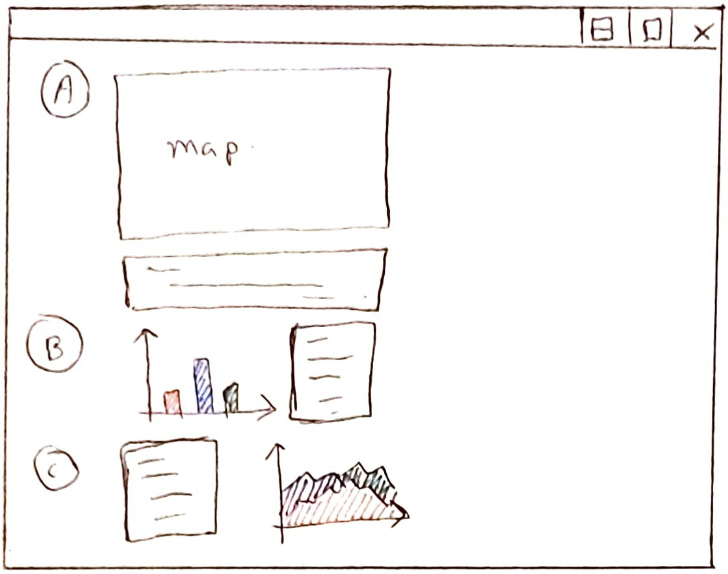
FILTER



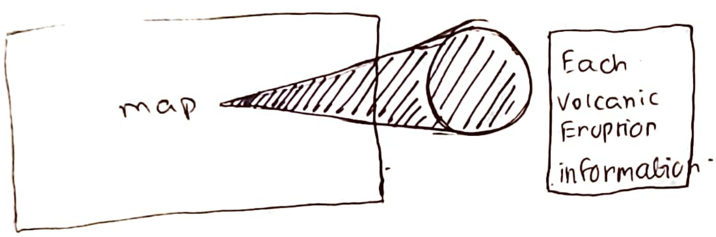
COMBINE AND REFINE

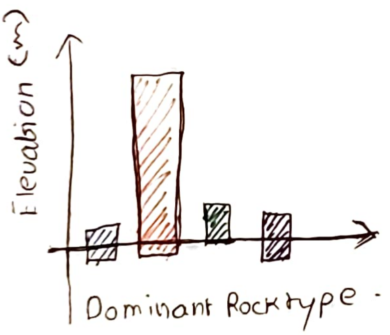
The idioms chosen will be interactive and will educate the viewers in an informative way. So it answers the original question

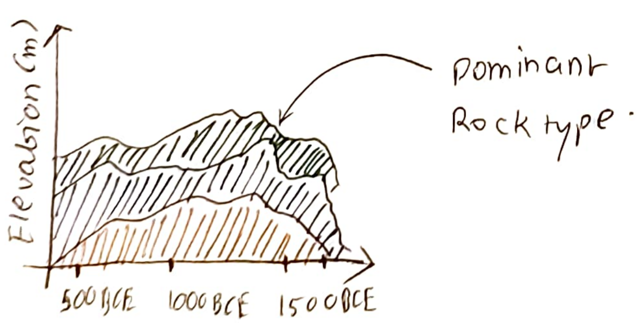
QUESTION



LAYOUT.







FOCUS :

Title : Visualisation project.

Author : Annur Mostafa

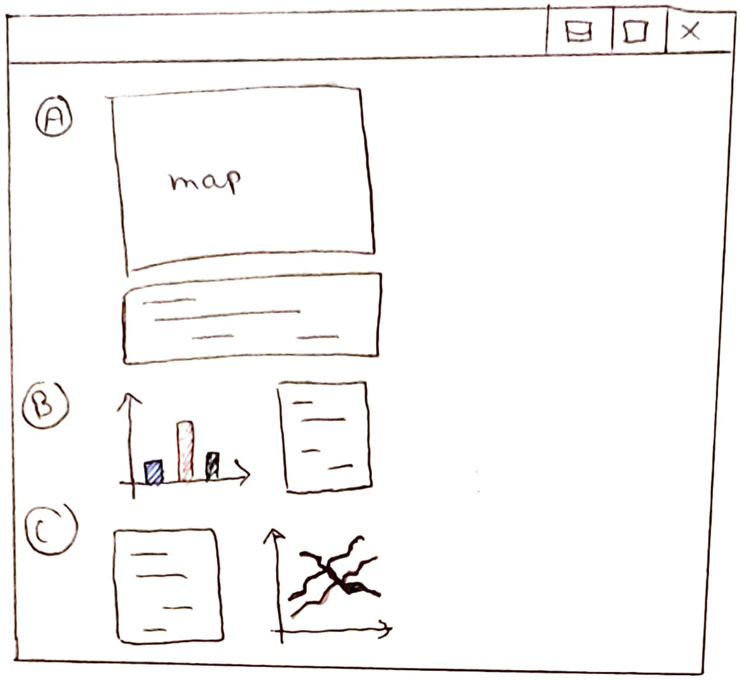
Date : 10 / 18 / 2023

Sheet : 02.

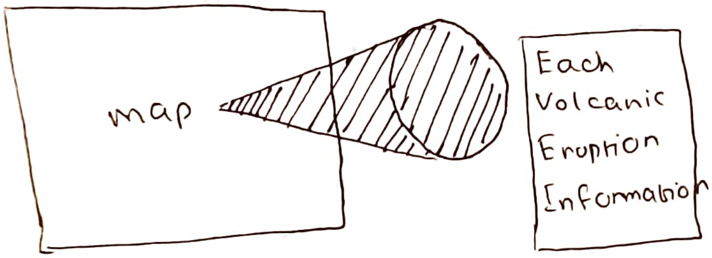
Task : Volcanic Eruption in the Holcene Period.

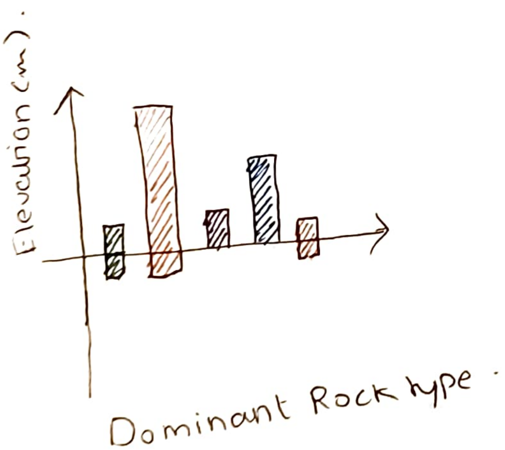
OPERATIONS AND DISCUSSIONS

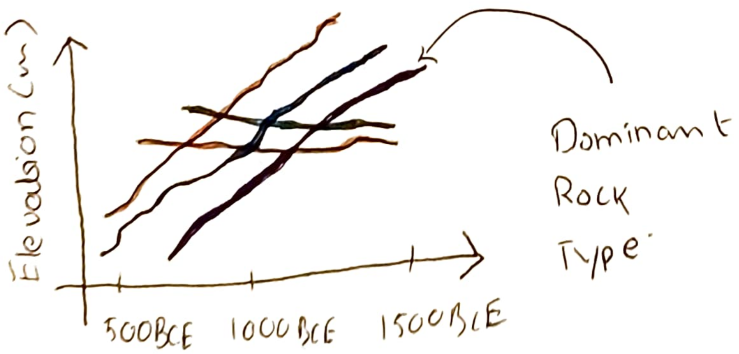
- The map is placed at the centre of the visual
- The map is interactive. The user can hover over each dot on the symbol map to see detailed information about each volcanic eruption on the map.
- The bar chart demonstrates negative as well as positive scale.
- Hovering over each area on the area chart shows each dominant rocktype.



LAYOUT







FOCUS

Title: Visualisation Project.

Author: Annur Mostafa

Date: 10/8/2023

Sheet: 03

Task: Volcanic Eruption in the Holcene Period.

Operations and Discussions

- The map is placed at the middle of the visualisation
- The map is interactive. The user can hover over each mark on the symbol map to see detailed information about each volcanic eruption on the map
- The bar chart demonstrates negative as well as positive y axis.
- Hovering over each line on the line chart shows each dominant rock type.

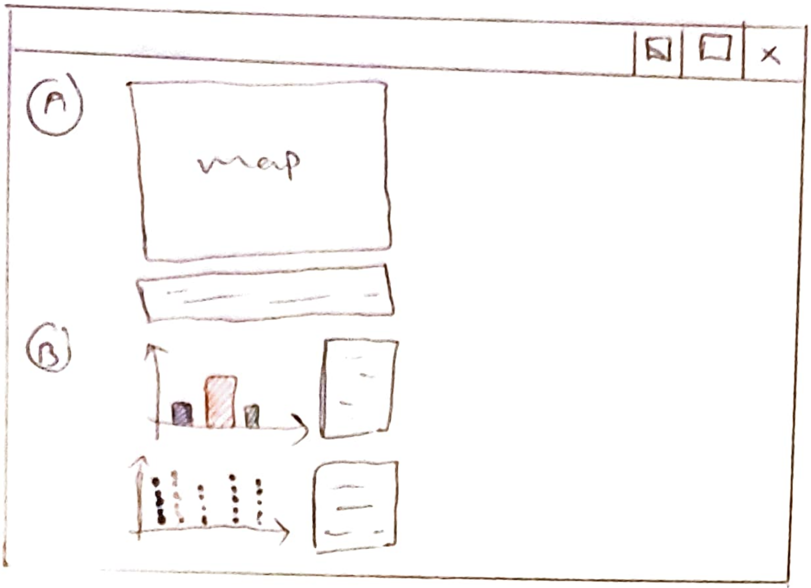
Title: Visualisation Project

Author: Annur mostafa

Date: 10/8/2023

Sheet: 04

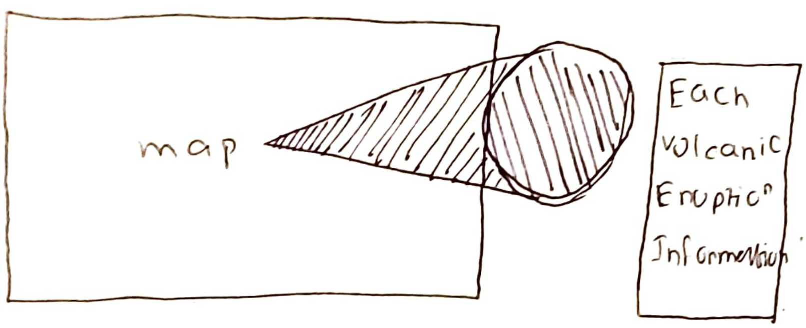
Task: Volcanic Eruption in the Holcene Period

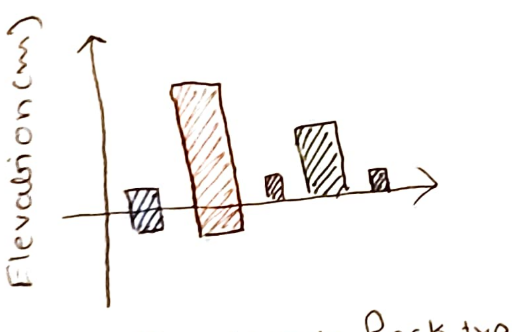


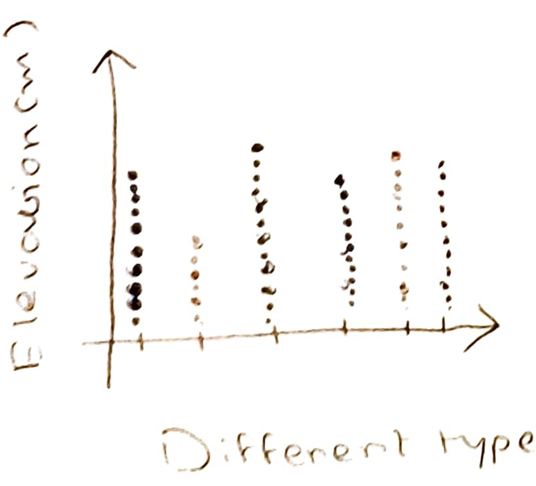
LAYOUT:

Operations and Discussions:

- The map is placed at the middle of the visualisation.
- The map is interactive. The user can hover over each mark on the symbol map to see detailed information about each volcanic eruption on the map.
- The bar chart demonstrates negative as well as positive y axis
-







Focus:

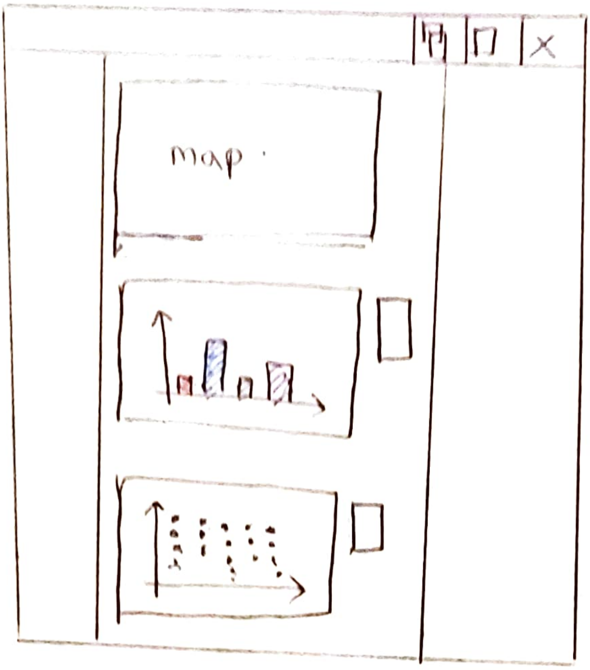
Title: Visualisation Project.

Author: Annur Mostafa

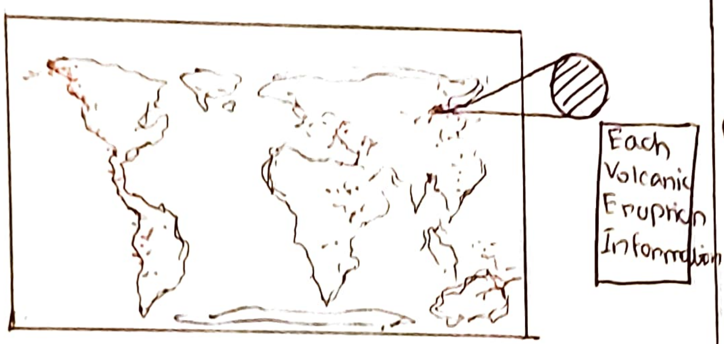
Date: 10/25/2023

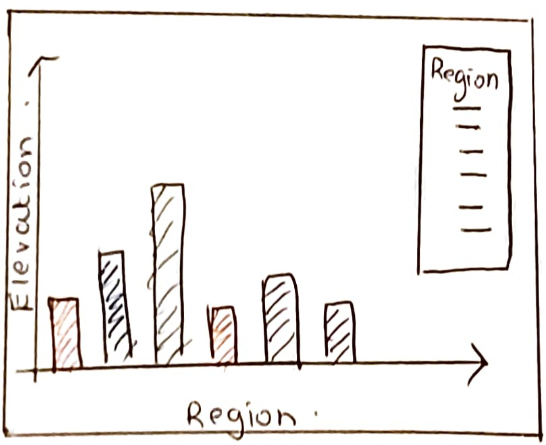
Task: Volcanic Eruption in the Holocene Period.

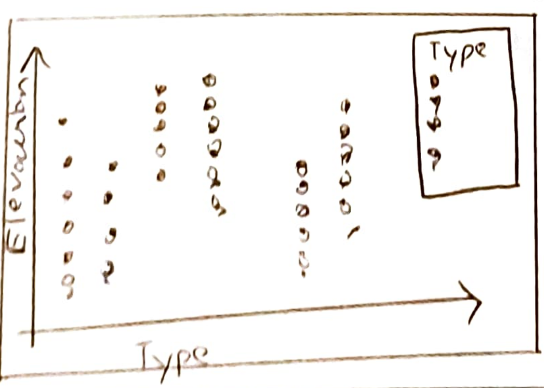
Sheet: 05.



LAYOUT







FOCUS

Operations AND Discussions.

1. The map is put at the middle of the visualisation
2. The map is interactive and when user move towards each mark, they can see detailed information about each volcanic eruption on the map.
3. The bar chart shows the highest and lowest elevations in each region
4. The scatter plot show the highest and lowest elevation of each type of eruptions.